

Tools of the Trade: Linux and SQL

Activity: Manage Files with Linux Commands

Introduction

In this activity, I practiced managing files and directories in Linux using the Bash terminal.

I created a new directory, removed an unused directory, moved and deleted files, and created a new file. I also used the `nano` text editor to add information to a file.

The main commands I used were `mkdir`, `rmdir`, `mv`, `rm`, `touch`, `nano`, `ls`, `cd`, and `cat`.

Activity Scenario

The task was to organize the `/home/analyst` directory.

I needed to create a `logs` directory, remove the unused `temp` directory, move `Q3patches.txt` from the `notes` directory to `reports`, and remove `temnotes.txt`.

After that, I created a new `tasks.txt` file in the `notes` directory and added a short note about the work I completed.

Lab Environment

- **Course:** Google Cybersecurity Certificate
 - **Course:** Tools of the Trade: Linux and SQL
 - **Activity:** Manage Files with Linux Commands
 - **Operating System:** Linux
 - **Shell:** Bash
 - **User:** `analyst`
-

Task 1: Create a New Directory

I started by creating the `logs` directory:

```
mkdir logs
```

I then checked the contents of the directory:

```
ls
```

The output showed:

```
logs notes reports temp
```

This confirmed that the `logs` directory was created successfully.

Task 2: Remove a Directory

The `temp` directory was no longer needed, so I removed it using:

```
rmdir temp
```

I checked the directory again:

```
ls
```

The output showed:

```
logs notes reports
```

The `temp` directory was successfully removed.

Task 3: Move a File

Next, I moved `Q3patches.txt` from the `notes` directory to the `reports` directory.

I first went to the `notes` directory:

```
cd /home/analyst/notes
```

Then I moved the file:

```
mv Q3patches.txt /home/analyst/reports/
```

I checked the `reports` directory:

```
ls /home/analyst/reports
```

The output showed:

```
Q1patches.txt Q2patches.txt Q3patches.txt
```

This confirmed that the file was moved successfully.

Task 4: Remove a File

I then removed the unused `tempnotes.txt` file:

```
rm tempnotes.txt
```

I used `ls` to check the directory:

```
ls
```

There were no files left in the `notes` directory at this point.

Task 5: Create a New File

I created a new file called `tasks.txt`:

```
touch tasks.txt
```

I checked that the file was created:

```
ls
```

The output showed:

```
tasks.txt
```

Task 6: Edit the File Using Nano

I opened the new file with the nano text editor:

```
nano tasks.txt
```

Inside the file, I added:

```
Completed tasks
```

1. Managed file structure in /home/analyst

I saved the file by pressing:

- CTRL + X
- Y
- ENTER

After exiting nano, I cleared the terminal:

```
clear
```

I then checked the contents of the file:

```
cat tasks.txt
```

The output was:

```
Completed tasks
1. Managed file structure in /home/analyst
```

This confirmed that the information was saved correctly.

Commands I Practiced

Command	What I used it for
mkdir	Create a directory
rmdir	Remove a directory
mv	Move a file
rm	Remove a file
touch	Create a new file
nano	Edit a file
ls	Check files and directories
cd	Move to another directory
cat	Read a file

What I Learned

This activity helped me practice basic file and directory management in Linux.

I learned how to create and remove directories, move and delete files, and create and edit a text file from the terminal.

I also became more familiar with the nano editor and learned how to save changes made to a file.

Cybersecurity Connection

File management is a basic but important part of working with Linux systems.

As a security analyst, I may need to work with log files, configuration files, reports, and other system files. Knowing how to find, move, create, remove, and read files from the command line will be useful when working with Linux systems and investigating security-related information.

Demonstration

The screenshots below show my practical work during the lab, including creating and removing directories, moving and deleting files, creating `tasks.txt`, editing it with nano, and checking the final contents.

```
analyst@4e1a5199e397:~$ tree
.
|-- notes
|   |-- Q3patches.txt
|   `-- tempnotes.txt
|-- reports
|   |-- Q1patches.txt
|   `-- Q2patches.txt
`-- temp

4 directories, 4 files
analyst@4e1a5199e397:~$ mkdir logs
analyst@4e1a5199e397:~$ ls
logs  notes  reports  temp
analyst@4e1a5199e397:~$ rmdir temp
analyst@4e1a5199e397:~$ ls
logs  notes  reports
analyst@4e1a5199e397:~$ cd /home/analyst/notes
analyst@4e1a5199e397:~/notes$ cd notes
-bash: cd: notes: No such file or directory
analyst@4e1a5199e397:~/notes$ mv Q3patches.txt /home/analyst/reports/
analyst@4e1a5199e397:~/notes$ ls /home/analyst/reports
Q1patches.txt  Q2patches.txt  Q3patches.txt
analyst@4e1a5199e397:~/notes$
```

Completed tasks

1. Managed file structure in `/home/analyst`

Completed tasks

1. Managed file structure in /home/analyst

```
analyst@4e1a5199e397:~/notes$ cat tasks.txt
Completed tasks

1. Managed file structure in /home/analyst
analyst@4e1a5199e397:~/notes$
```

Conclusion

In this activity, I practiced several Linux commands for managing files and directories.

I also used the nano editor to create and update a text file.

This was a useful exercise for building my Linux command-line skills, which are important for continuing my cybersecurity learning.

Course Information

Google Cybersecurity Certificate

Course: Tools of the Trade: Linux and SQL

Activity: Manage Files with Linux Commands

Disclaimer

This document is part of my personal cybersecurity learning portfolio. The activity was completed for educational purposes as part of the Google Cybersecurity Certificate.